

The degree-two case of the finite-free fourth-order comparison

Autonomous proof-discovery packet for arXiv:2602.10373

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Abstract

Heinävaara conjectures that the empirical root law of a finite free additive convolution is below the corresponding free convolution for every test function with nonnegative fourth derivative. We prove the conjecture for arbitrary monic real-rooted quadratic polynomials. The key observation is a general extremal principle: among symmetric compactly supported laws of fixed variance, the symmetric two-point law minimizes every fourth-order convex test. Its proof is a direct Jensen argument applied to the square of the random variable.

1 Source problem

Let p and q be monic real-rooted polynomials of the same degree d , and let μ_p, μ_q denote their empirical root measures. The finite free additive convolution $p \boxplus_d q$ is again real-rooted. Conjecture 5.2 of Heinävaara [1] asks whether

$$\frac{1}{d} \sum_{x: (p \boxplus_d q)(x)=0} f(x) \leq \int_{\mathbb{R}} f(t) d(\mu_p \boxplus \mu_q)(t) \quad (1)$$

for every $f \in C^4(\mathbb{R})$ satisfying $f^{(4)} \geq 0$.

The exact source passage is reproduced in Figure 1. It also records the weaker even-moment question that motivated the conjecture.

2 Result

Theorem 2.1 (Quadratic finite-free comparison). *Let p, q be arbitrary monic real-rooted quadratic polynomials. Then (1) holds with $d = 2$ for every $f \in C^4(\mathbb{R})$ with $f^{(4)} \geq 0$.*

In the notation for the fourth-order order introduced in the source, this is the assertion

$$\mu_{p \boxplus_2 q} \prec_4 \mu_p \boxplus \mu_q \quad (\deg p = \deg q = 2).$$

The result includes repeated roots. It is exact, not asymptotic, and the test class is the full class appearing in the conjecture.

Weak-* compactness of measures implies the existence of a limiting measure $\tilde{m}_{\mu,\nu}$ with the desired properties, at least for polynomial f . Approximating again deals with general $f \in C^4(\mathbb{R})$. $\square$

5. COMMENTS AND CONJECTURES

5.1. Finite free convolution and c-convolution. The author's original motivation was to understand a variant of inequality (3) that compares free convolution to *finite free convolution* of polynomials. Finite free convolution gained traction with the work of Marcus, Spielman, and Srivastava [MSS22]. The finite free additive convolution $p \boxplus_d q$ of two degree d polynomials with real roots is also a degree d polynomial with real roots. If μ_p and μ_q are probability measures with equal point masses at the roots of p and q respectively, the distribution $\mu_{p \boxplus_d q}$, for large d , is close to $\mu_p \boxplus \mu_q$ in a suitable sense [AP18, Corollary 5.3]. It follows from the work of Marcus, Spielman, and Srivastava [Sri24, Theorem 4.2] that the supports of finite free convolution and free convolution can in fact be compared,

$$\text{Conv}\{x \mid (p \boxplus_d q)(x) = 0\} \subset \text{Conv}(\text{supp}(\mu_p \boxplus \mu_q)).$$

Spielman allegedly asked whether this bound can be generalized to all even moments; namely, whether for $k \in \mathbb{N}$,

$$(50) \quad \frac{1}{d} \sum_{x: (p \boxplus_d q)(x) = 0} x^{2k} \leq m_{2k}(\mu_p \boxplus \mu_q).$$

While we were not able to answer this question, it lead to Theorems 1.3 and 1.4. It is conceivable that Theorem 1.3 (and 1.4) has a variant in the finite free setting.

Conjecture 5.2. *Let p and q be real-rooted polynomials, and $f \in C^4(\mathbb{R})$ with $f^{(4)} \geq 0$. Then,*

$$\frac{1}{d} \sum_{x: (p \boxplus_d q)(x) = 0} f(x) \leq \int f(t) d(\mu_p \boxplus \mu_q)(t).$$

It is possible to modify the arguments of section 3 to show that Conjecture 5.2 is true up to leading order, and the condition $f^{(4)} \geq 0$ is necessary.

Another avenue for generalization of our results is towards the so-called c -convolution of Mergny and Potters [MP22], $\mu \oplus_c \nu$, which convolves two compactly-supported probability measures μ and ν , and is parametrized by $c \in [0, \infty]$. Classical and free convolutions of μ and ν can be recovered as $\mu \oplus_0 \nu$ and $\mu \oplus_\infty \nu$, respectively. While c -convolution can be formally defined, either analytically

Figure 1: Conjecture 5.2 and its context on page 16 of arXiv:2602.10373.

3 Idea of the proof

Write the roots of p as $m \pm \alpha$ and those of q as $n \pm \beta$. The finite-free coefficient formula collapses to

$$(p \boxplus_2 q)(x) = (x - m - n)^2 - (\alpha^2 + \beta^2).$$

Thus, after removing the common center $m + n$, the finite root law is exactly the symmetric two-point law at $\pm r$, where $r^2 = \alpha^2 + \beta^2$.

The centered free convolution is the law of $Z = \alpha U + \beta V$, with U, V freely independent symmetric Bernoulli variables. It is symmetric and $\mathbb{E}Z^2 = r^2$. Fourth-order convex comparison is generated by the cubic stop-loss kernels $(x - t)_+^3$. For $t \geq 0$, symmetry gives

$$\mathbb{E}(Z - t)_+^3 = \frac{1}{2} \mathbb{E}(|Z| - t)_+^3.$$

The map $v \mapsto (\sqrt{v} - t)_+^3$ is convex, so Jensen compares the right-hand side with its value at $\mathbb{E}Z^2 = r^2$. This is precisely the cubic stop-loss transform of the two-point law. Symmetry handles negative t , and Taylor's integral remainder handles an arbitrary f .

4 A fourth-order extremal lemma

Lemma 4.1. *Let Z be a compactly supported symmetric real random variable and let $r^2 = \mathbb{E}Z^2$. Let Y take the two values $-r, r$ with probability $1/2$ each. Then for every $f \in C^4(\mathbb{R})$ with $f^{(4)} \geq 0$,*

$$\mathbb{E}f(Y) \leq \mathbb{E}f(Z).$$

Proof. For $t \in \mathbb{R}$, put

$$D(t) = \mathbb{E}(Z - t)_+^3 - \mathbb{E}(Y - t)_+^3.$$

First suppose $t \geq 0$. Define on $[0, \infty)$

$$g_t(v) = (\sqrt{v} - t)_+^3.$$

It vanishes on $[0, t^2]$. On (t^2, ∞) , direct differentiation gives

$$g_t''(v) = \frac{3(v - t^2)}{4v^{3/2}} \geq 0.$$

The value and first derivative glue continuously to zero at $v = t^2$; hence g_t is convex on all of $[0, \infty)$. By symmetry and Jensen's inequality,

$$\begin{aligned} \mathbb{E}(Z - t)_+^3 &= \frac{1}{2} \mathbb{E}(|Z| - t)_+^3 = \frac{1}{2} \mathbb{E}g_t(Z^2) \\ &\geq \frac{1}{2} g_t(\mathbb{E}Z^2) = \frac{1}{2} (r - t)_+^3 = \mathbb{E}(Y - t)_+^3. \end{aligned}$$

Thus $D(t) \geq 0$ for $t \geq 0$.

For a symmetric random variable W , symmetry and the elementary identity

$$(t - W)_+^3 - (W - t)_+^3 = (t - W)^3$$

give

$$\mathbb{E}(W + t)_+^3 = \mathbb{E}(W - t)_+^3 + \mathbb{E}(t - W)^3.$$

The variables Z and Y have the same moments through order three: both are symmetric, centered, and have second moment r^2 . Subtracting the last display for $W = Z$ and $W = Y$ yields $D(-t) = D(t)$. Therefore $D(t) \geq 0$ for every real t .

Choose a compact interval $[a, b]$ containing the supports of Z and Y . For $x \in [a, b]$, Taylor's formula with integral remainder gives

$$f(x) = P_3(x) + \frac{1}{6} \int_a^b (x - t)_+^3 f^{(4)}(t) dt,$$

where P_3 is a polynomial of degree at most three. The expectations of P_3 agree for Z and Y . Hence Fubini's theorem gives

$$\mathbb{E}f(Z) - \mathbb{E}f(Y) = \frac{1}{6} \int_a^b D(t) f^{(4)}(t) dt \geq 0.$$

□

5 Finite and free quadratic convolutions

Lemma 5.1. *Suppose*

$$p(x) = (x - m)^2 - \alpha^2, \quad q(x) = (x - n)^2 - \beta^2, \quad \alpha, \beta \geq 0.$$

Then

$$(p \boxplus_2 q)(x) = (x - m - n)^2 - \alpha^2 - \beta^2.$$

Proof. Write $p = x^2 - a_1x + a_2$ and $q = x^2 - b_1x + b_2$. Thus

$$a_1 = 2m, \quad a_2 = m^2 - \alpha^2, \quad b_1 = 2n, \quad b_2 = n^2 - \beta^2.$$

The degree-two specialization of the finite-free coefficient formula of Marcus–Spielman–Srivastava [2] is

$$p \boxplus_2 q = x^2 - (a_1 + b_1)x + \left(a_2 + b_2 + \frac{1}{2}a_1b_1 \right).$$

Substitution gives the claimed expression. $\square$

Lemma 5.2. *With the notation of Lemma 5.1, the free convolution $\mu_p \boxplus \mu_q$, translated by $-(m + n)$, is symmetric and has second moment $\alpha^2 + \beta^2$.*

Proof. Let U, V be freely independent symmetric Bernoulli variables, each taking the values $-1, 1$ with equal probability. The desired centered law is the law of

$$Z = \alpha U + \beta V.$$

The trace-preserving sign change $(U, V) \mapsto (-U, -V)$ in the free product shows that Z is symmetric. Also, freeness and centering give $\mathbb{E}(UV) = \mathbb{E}(VU) = 0$, while $U^2 = V^2 = 1$. Therefore

$$\mathbb{E}Z^2 = \alpha^2 + \beta^2.$$

$\square$

Proof of Theorem 2.1. Every monic real-rooted quadratic admits the parametrization in Lemma 5.1. Put $r^2 = \alpha^2 + \beta^2$. That lemma identifies the centered empirical root law of $p \boxplus_2 q$ with the symmetric two-point law at $\pm r$. Lemma 5.2 identifies the centered free convolution as a symmetric law with the same second moment. Apply Lemma 4.1, then translate both laws back by $m + n$. $\square$

Corollary 5.3 (Even moments in degree two). *For every integer $k \geq 1$,*

$$\frac{1}{2} \sum_{x: (p \boxplus_2 q)(x)=0} x^{2k} \leq m_{2k}(\mu_p \boxplus \mu_q).$$

Proof. For $k = 1$ there is equality by the first-three-moment calculation. For $k \geq 2$, take $f(x) = x^{2k}$ in Theorem 2.1; its fourth derivative is nonnegative on $\mathbb{R}$. $\square$

6 Verification, scope, and upgrade attempts

The symbolic verifier in `code/verify_degree_two.py` checks both algebraic identities on which the proof rests: the finite-free quadratic coefficient formula and

$$\frac{d^2}{dv^2}(\sqrt{v} - t)^3 = \frac{3(v - t^2)}{4v^{3/2}}.$$

The exploratory attempt also performed exact rational moment searches and Haar-matrix stop-loss searches; none is used in the proof.

Six materially different upgrade routes were pursued. The successful route above emerged after exact even-moment and stop-loss searches. For symmetric cubics the finite convolution has roots $0, \pm\sqrt{\alpha^2 + \beta^2}$, so its squared modulus is no longer constant and the Jensen extremality breaks. Existing finite-free majorization compares finite convolutions to other finite convolutions rather than to the free limit. Finally, the Mergny–Potters c -interpolation yields explicit hypergeometric laws only in special cases and did not provide a sign-definite stop-loss derivative.

Thus the packet settles the source conjecture completely in degree two but does not claim the general-degree result. Bounded exact-phrase and concept searches through 11 August 2026 found the source conjecture, finite-free cumulant papers, and the standard majorization results, but no prior statement of this quadratic fourth-order comparison. This is a novelty check, not a guarantee of novelty.

References

- [1] O. Heinävaara, *Convolution comparison measures*, arXiv:2602.10373, 2026.
- [2] A. W. Marcus, D. A. Spielman, and N. Srivastava, *Finite free convolutions of polynomials*, Probab. Theory Related Fields **182** (2022), 807–848; arXiv:1504.00350.
- [3] A. Nica and R. Speicher, *Lectures on the Combinatorics of Free Probability*, London Math. Soc. Lecture Note Ser. 335, Cambridge Univ. Press, 2006.